

Zhichao Wang

Curriculum Vitae

Email: zhw036@ucsd.edu

EDUCATION

University of California San Diego, USA

PhD candidate in Mathematics, 2019-present

- Advisors: Prof. Ioana Dumitriu and Prof. Todd Kemp

Texas A&M University, College Station, USA

MSc. in Mathematics, 2017-2019

- Advisors: Prof. Michael Anshelevich

Texas A&M University, College Station, USA

Exchange Program, 2016-2017

Beihang University, Beijing, China

BSc. in Mathematics and Applied Mathematics, 2013-2017

RESEARCH INTERESTS

- High-dimensional Probability · Random Matrix Theory and its Applications · Free Probability Theory
- Deep Learning Theory · Feature Learning · Kernel Methods · Random Feature Regression
- High-Dimensional Statistics · Asymptotic Statistics

PUBLICATIONS AND PREPRINTS

- **High-Dimensional Asymptotics of Feature Learning in the Early Phase of Neural Network Training** (with Jimmy Ba, Murat A. Erdogdu, Taiji Suzuki, Denny Wu, and Greg Yang), in preparation.
- **Overparameterized random feature regression with nearly orthogonal data** (with Yizhe Zhu), Accepted by AISTATS 2023.
- **Spectral evolution and invariance in linear-width neural networks** (with Andrew Engel, Anand Sarwate, Ioana Dumitriu, Tony Chiang), submitted.
- **High-dimensional Asymptotics of Feature Learning: How One Gradient Step Improves the Representation** (with Jimmy Ba, Murat A. Erdogdu, Taiji Suzuki, Denny Wu, and Greg Yang), Accepted by NeurIPS 2022.
- **Deformed semicircle law and concentration of nonlinear random matrices for ultra-wide neural networks** (with Yizhe Zhu), submitted.
- **Tree convolution for probability distributions with unbounded support** (with Ethan Davis and David Jekel), Latin American Journal of Probability and Mathematics Statistics 18.2 (2021), pp. 1585-1623.
- **Spectra of the Conjugate Kernel and Neural Tangent Kernel for linear-width neural networks** (with Zhou Fan), Neural Information Processing Systems (NeurIPS), 2020, *Oral Presentation*.
- **Principal components in linear mixed models with general bulk** (with Zhou Fan and Yi Sun), The Annals of Statistics, 49.3 (2021), pp. 1489-1513.
- **Higher variations for free Lévy processes** (with Michael Anshelevich), Studia Math. 252 (2020), pp. 49-81.
- **Convergence of the Powers of Free Triangular Arrays to Higher Variations of Free Lévy Processes**, Master's thesis, Texas A & M University, *available electronically*.

WORK EXPERIENCES

Internship at Pacific Northwest National Laboratory	<i>06/2021- 09/2022</i>
Advisor: Tony Chiang. Deep learning research and software development.	
Teaching Assistant at Math Department of UC San Diego	<i>09/2019-12/2021</i>
MATH 10B, MATH 3C, MATH 10C, MATH 180A, and MATH 280BC.	
Teaching Assistant at Math Department of Texas A&M University	<i>08/2018-07/2019</i>
MATH 251, MATH 308 and MATH 220.	
Grader of Math Department at Texas A&M University	<i>01/2017-05/2018</i>
MATH 467, MATH 411 and MATH 152.	
Internship at AMSS, Chinese Academy of Sciences	<i>07/2016-08/2016</i>
Advisor: Prof. Feimin Huang, AMSS, Chinese Academy of Sciences	
Mentor at Math Department, Beihang University	<i>10/2014-07/2015</i>

HONORS AND AWARDS

<i>2019</i>	Math Fellowship, UCSD, USA
<i>2017</i>	Math Fellowship, TAMU, USA
<i>2016, 2017</i>	Study Abroad Scholarships, Beihang University, China
<i>2016</i>	Meritorious Winner of MCM/ICM Contest, USA
<i>2016</i>	Second Prize in the 26th Fengru Cup Competition of Academic Research Works, China
<i>2015, 2016</i>	Hua Luogeng Scholarships, AMSS, Chinese Academy of Sciences
<i>2014, 2015</i>	Huatong Scholarships of SMSS, Beihang University, China
<i>2014, 2015</i>	Scholarships of Academic Performance, Beihang University, China
<i>2014</i>	First Prize in College Students Physics Contest in Beijing, China
<i>2013</i>	Freshman Scholarship, Beihang University, China

TALKS

<i>06/2022</i>	RMMC Summer School at University of Wyoming.
<i>03/2022</i>	Combinatorics and Probability Seminar at UC Irvine.
<i>12/2021</i>	Frontier Probability Days 2021, at Las Vegas.
<i>09/2021</i>	Universality and Integrability in Random Matrix Theory and Interacting Particle Systems Workshop, at MSRI (virtual).
<i>03/2021</i>	Machine Learning Seminar, at Pacific Northwest National Laboratory (virtual).
<i>12/2020</i>	Neural Information Processing Systems (NeurIPS) virtual oral presentation.
<i>03/2020</i>	Student Probability Seminar, at UCSD.
<i>11/2019</i>	Student Probability Seminar, at UCSD.
<i>05/2017</i>	Capstone Project Presentation, TAMU.
<i>04/2016</i>	Complex Dynamic Systems Seminar, Beihang.

SKILLS

<i>Language</i>	Chinese (Native), English (Fluent)
<i>Software</i>	Python, MATLAB, L ^A T _E X, Tensorflow, Keras, PyTorch, JAX, C++

PROFESSIONAL SERVICE

Journal reviewer: Annals of Statistics, Canadian Journal of Mathematics.
Conference reviewer: AISTATS '22, AISTATS '23.